Results and Evaluation

> Draft of the paper's Results & Evaluation section. Written in the paper's framing (a computer-science / software-engineering framework; see PAPER_FRAMING.md): we evaluate the framework's behaviour — how it wraps an unreliable AI component into a dependable, extensible system — not clinical diagnostic accuracy. Every figure below is the value the current system produces on the demonstration dataset; regenerate with PYTHONPATH=. python tests/evaluate_<name>.py --write. Numbers are reported as engineering quantities measured against a reference severity ordering, not as medical claims.

5.0 Experimental setup

Demonstration dataset. All experiments use a public 41-disease symptom dataset (4,920 rows; 304 unique symptom profiles) as a controlled test-bench for exercising the framework. Its severity tiers (EMERGENCY/SEVERE/MODERATE/MILD) are an author-assigned reference ordering used to measure framework behaviour, not clinical ground truth; the dataset is synthetic and not representative of real rural presentations. We treat this as a sandbox to show the system behaves as designed, and state clinical validation as future work.

Protocol. Unless noted, evaluation is out-of-sample (leave-profiles-out 5-fold: every test profile is scored by a model that never trained on it) and on short messages (each profile truncated to 2–4 symptoms), which is the realistic first-contact condition — a caregiver reports a few symptoms, not a full 17-symptom record. Token-scored experiments are deterministic (fixed seed); language and reliability experiments use a local `llama3.2:3b` and a multilingual sentence-embedding matcher.

Reporting conventions. We report two behavioural safety quantities in plain terms: "misses-urgent" (a case whose reference tier is EMERGENCY/SEVERE surfaced at a lower tier) and "over-calls" (a MILD/MODERATE case surfaced as EMERGENCY/SEVERE). We also report the decision mix (confident / uncertain / abstain) and, where relevant, reproducibility and cost.

Reproducibility. The system ships 627 automated tests (CI-gated) and deterministic evaluation harnesses that write tracked baseline JSON files; the deterministic harnesses reproduce byte-for-byte. The results below are drawn from those baselines.

5.1 A single AI model is not sufficient (motivation)

The natural challenge to any framework of this kind is that a modern LLM might do the task directly. We test this: `llama3.2:3b` is asked to assign the same four tiers on the same held-out cases, with the same information and a prompt that states the tier definitions and the safety convention ("when uncertain, choose the more severe tier") — so the baseline is not a straw man.

System	misses-urgent	over-calls	exact-tier accuracy
LLM-only (`llama3.2:3b`)	0.169	0.827	0.454

The model is simultaneously unsafe and uninformative: it lets 16.9 % of serious cases through at a lower tier while over-calling 82.7 % of mild/moderate ones, because it collapses onto a single answer — 251 of 304 replies are "SEVERE". Prompted safety instructions do not produce usable triage behaviour. This is the motivation for the framework: the contribution is the software structure that turns this component into a dependable system.

5.2 The framework, component by component

5.2.1 Composed ablation

We walk a single behavioural metric across the whole stack on the same held-out short messages, adding one mechanism at a time (`tests/evaluate_composed.py`):

Stage	Configuration	misses-urgent	over-calls
C0	LLM-only (prompt the model)	0.169	0.827
C1	+ structured classifier (argmax)	0.092	0.068
C2	+ abstain gate (uncertainty set)	0.063	0.068
C3	+ worst-case guard (full system)	0.000	0.216
C4	+ active follow-up (shipped)	0.021	0.056

Two things are visible in one table. First, structure alone (C1) already removes most of the LLM's pathology — misses-urgent falls 0.169 → 0.092 and over-calls 0.827 → 0.068. Second, the worst-case guard (C3) is the mechanism that drives misses-urgent to zero, at the honest cost of over-calling (0.216); the active follow-up loop (C4) then buys most of that cost back (over-calls 0.216 → 0.056) while keeping misses-urgent near zero (0.021).

5.2.2 Failure modes of naive AI use, and the design response

The framework is best understood as a set of mitigations, each removing a concrete, observed failure of the naive approach. Every row below is a defect we measured in the system and the design decision that closes it.

Observed failure of naive AI use	Design response	Measured effect
LLM mode-collapse → everything "SEVERE"	structured classifier + abstain gate	over-calls 0.827 → 0.216
Fragile extraction — one schema-handling bug discarded whole cases	schema repair + input grounding	extraction failure 13.3 % → 0.0 %; expected-token recall 80.0 % → 100 %
Intensity words corrupting a symptom ("mild fever" → the wrong token)	extraction-prompt fix	corrected (regression-locked)
One-word messages dropped entirely	prompt + raw-text fallback	corrected
Hallucinated diagnosis leaking into a clinical handoff (a convulsing infant mislabelled with an unrelated chronic disease and its home-care advice)	suppress the supplementary diagnosis on danger-sign / abstaining paths	leak removed
Incoherent follow-up questions put to a layperson (e.g. asking about "acute liver failure")	restrict questions to answerable, characteristic symptoms	unanswerable-question rate 53.6 % → 0.0 %
Negative evidence flattening the classifier → a 28-disease set wrongly escalated to EMERGENCY	restrict absence evidence to the supported differential	escalation impossible (invariant-tested)

5.2.3 The safety property is a guarantee, stated honestly

The framework labels an uncertainty set by its most severe member. Whenever the true tier is in the set, the surfaced tier is therefore at least as severe as the truth, so a miss can only happen when the set omits the truth — an event the uncertainty gate bounds. In plain terms, the system's miss rate is bounded by a single tunable parameter (the gate's miscoverage budget, α): on the held-out set, coverage is 1.0 and misses-urgent is 0.000, with the price paid as over-calls (0.216). We present this as a design guarantee with a measured premise, not as an empirical surprise.

We report two honest disclosures a reviewer would otherwise find. (i) At the shipped setting the gate is conservative: on 33.9 % of cases its set is empty and the system returns all surviving candidates, so part of the coverage is a fallback rather than selective filtering. (ii) A parameter sweep shows the bound holds across the

whole range of α and that a tighter setting (α = 0.01) is at least as safe with less over-calling and no fallback — so α is reported as a tunable dial, and a secondary emergency-class weighting is reported as a negative result (it changes neither safety quantity and is a candidate for removal). The statistical machinery behind the gate is an implementation choice, detailed in the appendix; the contribution is its role in the framework.

The full-system decision mix on the held-out short messages is 201 confident / 55 uncertain / 48 abstain (of 304), and the tier is stable across two different symptom subsets of the same patient 86.6 % of the time.

5.3 Active follow-up: converting uncertainty into action

The follow-up loop is the framework's novel coordination mechanism: when the classifier is unsure, the system asks the caller a small number of targeted yes/no questions and re-decides. We evaluate it out-of-sample with a 3-question budget; answers are read from each held-out patient's real recorded symptom profile — a ground-truth lookup, not a simulation (`tests/evaluate_followup.py`).

Evidence model	Question policy	resolved	mean Q	set before → after
ignores "no" (early)	information-gain selector	0.194	—	4.33 → …
ignores "no" (early)	random relevant question	0.291	—	—
counts "no" (shipped)	information-gain selector	0.748	2.15	4.33 → 1.67
counts "no" (shipped)	random relevant question	0.583	—	—

This is a two-part finding and the paper's most interesting result. Scored on positive evidence only, the "clever" selector is worse than asking a random relevant question (0.194 vs 0.291): its objective models the value of a "no", but the classifier discarded every "no", so half the information each question was chosen to obtain was thrown away — a genuine design inconsistency we found by measuring. Making a "no" real evidence raises resolution 0.194 → 0.748 (≈ 3.9×) and the selector then dominates both baselines, resolving three-quarters of uncertain cases in about two questions.

Shipping that fix surfaced a second lesson worth reporting. The obvious implementation (apply the absence term across all diseases) let a few "no"s flatten the classifier, producing a 28-disease set that the worst-case guard then escalated to EMERGENCY — denials made the system less certain and more alarmed. Restricting absence evidence to the differential the positive symptoms already support removes this: a "no" can now only narrow or re-rank, never introduce an unsupported disease or raise the tier (asserted by test). The loop's net effect on the full system is favourable: over-calls fall 0.216 → 0.056 for 0.021 misses-urgent (within the bound), with the true tier retained in the set 97.4 % of the time.

5.4 Extensibility: a second agent on the same framework

A framework earns the name only if others can extend it. We demonstrate this by building a second, different agent — outbreak surveillance (Agent 2) — on the same structure and showing how little it touches.

Coupling (measured, test-enforced). Agent 2 consumes the same case record the triage pipeline already writes, through a minimal `all_cases()` interface, and imports none of the triage decision code — a structural test fails the build if it ever does. The cost to add it is a new self-contained package, one UI view, and one additive database column, with zero edits to the triage decision path. The model backend is likewise a genuine swap point (three interchangeable implementations behind one interface).

Behaviour (demonstration stream). Agent 2 reads the case stream, flags unusual disease clusters per village with simple baseline-deviation statistics, and — matching its design intent — raises an alert only when something is out of hand, autonomously writing the official a natural-language situational brief and a

recommended public-health action. On a synthetic multi-village stream with one injected outbreak it surfaces exactly that outbreak (a diarrhoeal-disease cluster, 12 cases this week versus a baseline of 0, concentrated in under-5s) as a single alert and stays silent on a matched flat stream (0 alerts). The LLM is invoked only when an alert fires, so a quiet region costs no model calls. An optional feedback path (an active outbreak raising the triage prior for that disease in that area) demonstrates bidirectional coordination between agents; it ships disabled by default so the triage core is unaffected.

5.5 Practical and SDG-relevant impact (with vs without)

These results translate framework behaviour into terms relevant to the use case and to UN SDG-3 (access to health services) and SDG-10 (reduced inequalities). They are demonstrated on the sandbox dataset; the cost figures are an explicitly illustrative model with stated assumptions, not field measurements (tests/evaluate_impact.py).

A. Access and avoided journeys. The system turns 42.1 % of messages into a clear "care at home" decision (potential avoided trips) and 57.9 % into "seek care". For the latter it names the nearest eligible facility: without eligibility filtering, the naive "go to the nearest clinic" choice sends an emergency to a facility that cannot treat it in 80 % of the villages in the demonstration network. Under a stated model (one avoided trip ≈ 4 h and ≈ ☐300 per caregiver), the care-at-home share corresponds to ≈ 421 avoided trips and ≈ 1,684 hours per 1,000 callers — reported as an illustration of scale, not a measurement.

B. Language reach and inclusion. The pipeline serves English, Hindi and Hinglish: on real code-mixed lay phrasings it recovers 96.2 % of the intended symptoms, gives the same decision to repeated messages (1.00) and to paraphrases (0.75), and never mis-reads or coerces a hedged reply (0.0 each). An English-only pipeline would simply drop the non-English share of callers.

C. Caller effort. Reaching a decision costs the caregiver about two yes/no questions (mean 2.15), resolving 74.8 % of uncertain cases — rather than a full structured form or an unsafe guess.

5.6 Supporting results

End-to-end (free text in, tier out). Running the whole pipeline — LLM extraction and symptom matching included — on rendered messages, misses-urgent is 0.000 both on canonical phrasings (token recall 0.993, 99.7 % agreement with an oracle on clean tokens) and on real code-mixed lay phrasings (token recall 0.962, 97.6 % agreement); extraction costs precision (a few points of over-calling), not safety.

Robustness and reply handling. Identical messages give an identical decision (1.00); three real paraphrases of the same content agree 0.75 of the time; across 48 documented reply forms, none is misunderstood or mis-read and no hedged reply ("maybe", "pata nahi") is coerced into an answer (so the caller is re-prompted rather than having a fact invented for them).

Prioritisation and routing. The within-tier urgency score orders two-thirds of same-tier case pairs (discrimination 0.659), i.e. it carries information the tier label does not; the router's choice matches an exhaustive optimum (1.00) and always respects emergency-eligibility and opening hours on the demonstration network (mean ETA ≈ 12 min). No clinical-correctness claim is attached to these.

Systems cost. The framework is deliberately light: exactly one LLM call per case is unavoidable (extraction, ≈ 3.8 s on the test machine); every other model call is conditional, and surveillance's brief is event-driven. The deterministic stages are negligible by comparison (routing ≈ 0.06 ms, a full surveillance run ≈ 0.23 ms, classification a few ms per case), so the framework adds essentially no compute over the single model call it wraps.

Reproducibility. 627 automated tests pass (1 expected-fail, 1 expected-pass); the deterministic harnesses reproduce byte-for-byte and write the tracked baseline files these numbers are read from.

5.7 Threats to validity

We state the scope plainly. The data is a synthetic demonstration dataset: its severity labels are an author-assigned reference ordering, only two of its diseases are EMERGENCY, and it is not representative of real presentations, so the absolute rates should be read as framework behaviour on a test-bench, not as clinical performance. The access/cost figures are an illustrative model with stated assumptions. There is no user study or field deployment, and the realistic rural channel (SMS/IVR) is named as future work. The follow-up results have limited headroom because the dataset's symptom–disease associations are near-binary, and the safety bound becomes approximate once adaptively collected "no" answers are present (verified empirically to still hold: post-follow-up misses-urgent 0.021 ≤ the budget). These are expanded in the Limitations section; the system makes no clinical diagnostic-accuracy claim.